

Kartheeswaran Rajasekaran

📍 Coimbatore, Tamil Nadu, India

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Summary

- Software Engineer with 2.5+ years of combined professional and internship experience designing scalable, high-performance web applications using ASP.NET Core, Django REST Framework, and Express.js.
- Experienced in secure API architecture (JWT, OAuth2, RBAC), cloud-native deployments on Azure and VPS environments, database optimization, CI/CD automation, and real-time UAV system integrations.

Experience

LKPR Global

Jul 2025 – Present

Software Developer

Tamil Nadu, India

- Architect scalable full-stack platforms using Express.js, React.js, and MySQL supporting 10,000+ users with 99.5% production uptime.
- Engineered 15+ secure REST APIs with JWT authentication, RBAC, validation middleware, and pagination, reducing security vulnerabilities by 35%.
- Optimized database indexing and query execution plans, improving API response performance by 40% under concurrent traffic.
- Orchestrated CI/CD workflows via GitHub Actions and containerized deployments, accelerating release cycles by 30%.

Trilochana Aerospace & Defence Technologies

Jul 2024 – May 2025

Software Engineer Intern

Remote

- Engineered real-time UAV monitoring backend using Django REST Framework and MongoDB achieving <100ms latency for telemetry streaming.
- Integrated MAVLink protocol with Pixhawk drones, enhancing telemetry reliability and flight stability by 15%.
- Enabled concurrent drone session handling through WebSocket communication and asynchronous request processing.

Abstract Tech Groups

Feb 2022 – Jul 2023

Backend Software Developer

Remote, UK

- Designed enterprise ASP.NET Core MVC systems supporting 100,000+ daily transactions for UK government platforms.
- Refactored LINQ queries and ORM access layers, improving backend throughput and reducing database load by 25%.
- Provisioned Azure App Services, Azure SQL, and Blob Storage ensuring secure and high-availability deployments.
- Strengthened authentication mechanisms using OAuth2 and Azure AD SSO, elevating access compliance standards.

Skills

Languages: C#, Python, JavaScript (ES6+), TypeScript, SQL

Frameworks: ASP.NET (Web API, MVC), ADO.NET, Django REST Framework, Express.js, React.js, Next.js, ShadcnUI, xUnit

Databases: Microsoft SQL Server, MySQL, PostgreSQL, MongoDB

Cloud & DevOps: Microsoft Azure, GitHub Actions, CI/CD, Docker, VPS Deployment

Authentication: JWT, OAuth2, Azure AD, RBAC

Concepts: REST APIs, Linq, Microservices, WebSockets, Database Indexing, Query Caching, Agile

Education

Kumaraguru College of Technology <i>Master of Technology in Defence Technology (CGPA: 8.1 / 10)</i> • Relevant Coursework: Computational Fluid Dynamics, Drone Design, Autonomous Systems & Navigation Systems, Embedded Softwares for flight controls.	Aug 2023 – May 2025 Coimbatore, Tamil Nadu
PSR Engineering College <i>Bachelor of Engineering in Computer Science & Engineering (CGPA: 8.2 / 10)</i> • Relevant Coursework: Data Structures, Algorithms, Networking & Routing, Cloud Services, Python Programming, Web & App Development.	Sept 2018 – May 2022 Sivakasi, Tamil Nadu

Projects

Political Party Management Portal <i>React.js, Express.js, MySQL, JWT, VPS Deployment</i> • Designed backend services for structured party data management with JWT authentication, RBAC, validation middleware, pagination, and secure image uploads. • Optimized REST endpoints and event scheduling modules, improving operational efficiency by 30% across cloud deployments (Hostinger + Vercel).	Oct 2025 – Jan 2026
Tamilarasu Government E-Magazine Portal <i>PHP, JavaScript, MySQL, Razorpay API</i> • Architected secure e-commerce backend for government e-book purchases integrating Razorpay payment gateway. • Improved read performance by 35% using caching and SQL optimization while maintaining scalable VPS deployment.	Jul 2025 – Oct 2025
Universal Drone Monitoring & Maneuvering System <i>Django REST Framework, MongoDB, WebSockets, MAVLink</i> • Built real-time UAV telemetry platform supporting concurrent drone sessions with <100ms API latency. • Enhanced communication stability by 20% using asynchronous processing and WebSocket streaming.	Jul 2024 – May 2025
Digital Signage Platform <i>.NET Core, React.js, Azure AD, OAuth2</i> • Developed scalable multimedia content APIs integrated with Azure AD SSO authentication. • Increased deployment reliability by 20% across multi-tenant Azure cloud environments.	Sept 2021 – Dec 2022

Certification Courses

Introduction to Generative AI <i>Google</i>	Mar 2025
AWS Solution Architect <i>Amazon Web Services</i>	Dec 2021
Network Virtualization & Simulation <i>VMware</i>	Aug 2021
Deep Learning & Natural Language Processing <i>MathWorks Academy</i>	Aug 2020
CCNA – Routing and Switching <i>Cisco Networking Academy</i>	Apr 2020

Interests

- Developing full-stack, AI-enabled applications integrating secure backend architectures with real-time data processing and interactive frontend systems.
- Exploring autonomous drone systems, computer vision, cloud-native deployments, and scalable distributed system design.